

6G at 1/6G: A Review and Future Directions for Cislunar Communications

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ABSTRACT

What will the future of cislunar communications be? The ever-expanding horizons of space exploration missions and the need to establish sustainable space communication and navigation infrastructure necessitate thinking about this question thoroughly. On the other hand, steady progress in terrestrial communication leading up to 5G and 6G networks has presented a variety of ideas and optimization techniques. In this article, we explore the potential for improvement in cislunar communications with the backdrop of new technological advancements. In the article, we discuss ongoing cislunar developments, goals for the near future, and 6G concepts, such as reconfigurable intelligent surfaces, quantum-resistant physical layer security, private information read/write/cache networks, semantic and goal-oriented communications, information freshness-based quality of communication metrics, and multi-relay and cooperative networks. In addition, we present relevant promising avenues to explore to shape the future of cislunar communications.

INTRODUCTION

Cislunar communication plays a pivotal role in the progress of human civilization toward space exploration by facilitating successful missions and broadening the reach of technology. As the complexity of lunar missions has gradually increased over time, the characteristics of cislunar communications have also progressed accordingly. From early robotic space explorations to crewed Apollo missions to the currently ongoing efforts of building sustainable human bases on the lunar surface, the ever-growing development of missions necessitates robust, reliable, and efficient cislunar communication protocols.

The term “cislunar space” is used to refer to the volume between the Earth and the moon. The average distance between our planet and its natural satellite is 384,400 km, which, to put into perspective, is approximately 40-thousand times the altitude at which passenger planes fly. A signal propagating at the speed of light takes around 1.3 seconds to cover that distance, establishing a lower bound on the communication delay and significantly impacting the design of network protocols, as discussed later.

The cislunar space (Fig. 1) can be characterized by five Lagrangian (libration) points (denoted by L_1 to L_5) which are unique points in space

where the gravitational attraction force from the Earth and the moon interact to create an equilibrium. These points remain stationary relative to the Earth and the moon, and this enables objects to attain various stable orbits in the cislunar space. One such orbit is the halo orbit around the L_2 Lagrangian point depicted in purple in Fig. 1. Distant retrograde orbits (DRO), illustrated in red, are another special class of orbits whose interactions with the L_1 and L_2 Lagrangian points make them highly stable. DRO objects orbit around the Earth; however, relative to the moon, they form quasi-elliptic orbits that rotate in the direction opposite to the direction in which the moon orbits the Earth. The existence of these stable orbits is instrumental for the deployment of the necessary infrastructure for cislunar communications.

In parallel to the development of cislunar communications, there has been steady progress in terrestrial communications, the most recent version of the communication standard being the fifth generation (5G) communication. There has been commercial deployment of cellular-to-satellite links for text messages [1]. The ongoing research work on the next generation, that is, 6G communications, shows promising aspects of communication that were not considered in previous generations. Examples of such innovative directions include multi-relay transmission [2], semantic and goal-oriented communications [3], freshness and timeliness of information as an emerging quality metric [4], reconfigurable intelligent surfaces [5], use of information-theoretic security and privacy measures, such as physical layer security [6] and private information read, write, cache. In this article, we consider different insights from these emerging fields in the context of cislunar communication and provide plausible future directions of how they can be adopted for further improvement.

VISION FOR CISLUNAR SPACE: LUNANET

Historically, lunar missions were sparse and isolated, with significant gaps between successful landings. This trend is observed from the Luna 9 mission in 1966 to the more recent Chandrayaan-3 in 2023. However, the surge in planned missions, particularly those targeting the lunar south pole for its valuable resources, necessitates a robust, scalable, and interoperable communications and navigation (C&N) infrastructure to sup-

port long-term science, exploration, and industrial activities on the moon. As outlined in the space communications and navigation (SCaN) program [7], the future of cislunar infrastructure relies on collaboration between government agencies and commercial and international partners.

The main backbone of this vision is the LunaNet architecture introduced in [8]. LunaNet is a flexible and extensible C&N architecture designed to ensure interoperability among various stakeholders. Creating an interoperable global lunar utilization infrastructure is essential, as this system will allow US industry and international partners to maintain a continuous presence on the moon without NASA being the sole user. LunaNet aims to provide key services such as machine-to-machine (M2M) communication, networking, position, navigation, and timing (PNT). The node-based architecture of LunaNet enables it to cater to a wide range of applications. Depending on the requirements and constraints of these applications, the nodes of the LunaNet may be arranged in a variety of configurations, such as node clusters sharing a single direct link to the Earth (Fig. 2a) or nodes communicating via relay stations (Fig. 2b). The development of LunaNet involves close collaboration with industry partners to refine and implement interoperability specifications, and to ensure that the system can support a diverse range of lunar activities, such as autonomy for lunar robotics, and services for robotic and human missions.

The SCaN program [7] outlines a planned approach to developing a cislunar infrastructure. In the initial stage, the plan is to upgrade the existing deep space network (DSN) antennas and the development of new 18-meter antenna subnets (LEGS) to enhance lunar communication capabilities. This includes increasing data rates and supporting simultaneous operations across multiple frequency bands. Consequently, relay services will be established to ensure continuous coverage, particularly at the lunar south pole and far-side regions. The final goal includes the implementation of 3GPP-based cellular technology, optical communications for high bandwidth needs, and the development of lunar navigation services similar to the GPS on Earth for the moon. With this backdrop, we will explore some emerging 5G and 6G technologies in later sections. Even though those technologies might not be readily extended to the cislunar setting due to different communication environments and resource limitations, the ideas can be helpful to continue further research to facilitate the project deliverables of the program more efficiently.

In [7], the project deliverables are outlined in the following three phases. The *initial phase* (2024–2025) includes establishing foundational PNT communication and navigation services, including direct-to-earth (DTE) and initial relay services. The *growth phase* (2026–2030) includes expanding relay services, introducing optical links, and enhancing surface networking capabilities. Finally, the *desired future state* (2030+) includes achieving a comprehensive and resilient infrastructure with satellite constellations, and extensive surface networks.

The success of the cislunar infrastructure depends heavily on international partnerships and contributions. The interagency operations advisory group (IOAG) and the international communication system interoperability standards (ICSIS)

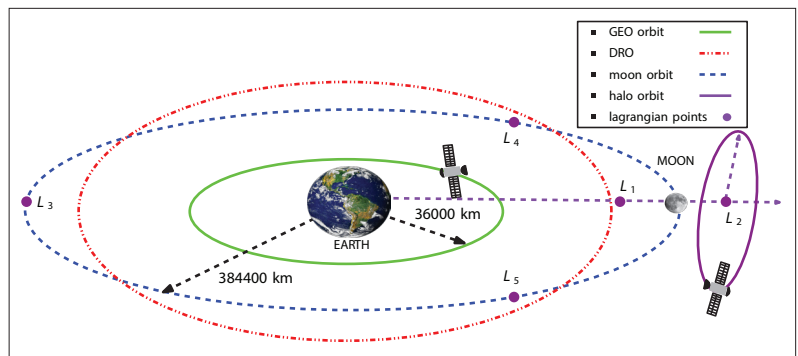


FIGURE 1. The cislunar space illustrating several significant orbits.

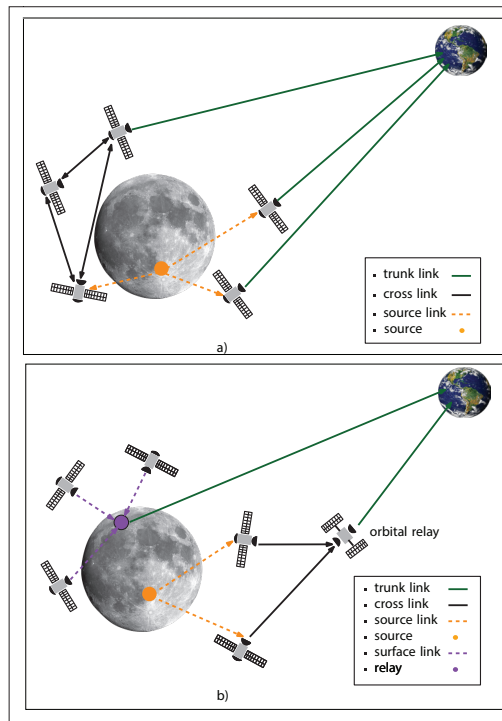


FIGURE 2. Variations of the LunaNet node configurations: a) Node clusters connected via cross-links sharing a single trunk link and isolated nodes with dedicated trunk links; b) Nodes communicating via a lunar surface relay or a high lunar orbit relay station.

will play crucial roles in coordinating efforts and ensuring that systems developed by different entities can work seamlessly together.

EMERGING TECHNOLOGIES AND CONCEPTS

Humankind has set its sights on establishing a long-term presence on the moon to facilitate deep space missions and the colonization of Mars. This endeavor cannot be successful unless reliable, high-rate communication in the cislunar space is available. With the increasing number of planned moon missions in the last few years, the landscape of communication is expected to be very different from that of the first crewed mission to the moon, necessitating the development of new communication architectures and technologies.

OPTICAL COMMUNICATIONS

One of the key technologies behind high-rate communications in the cislunar space is free space optical communications. Not only do they provide significantly higher data rates compared

Link	Transmitter Aperture – Power	Receiver Aperture (cm)	Data Rate (Gb/s)	Distance (km)	Propagation delay (sec.)
Moon-Earth Trunk	10 cm – 10 W	100	5	385,000	1.3
Spacecraft-Relay	10 cm – 1 W	10	2	70,000	0.23
Lunar User-Relay	5 cm – 1 W	10	0.2	10,000	0.03

TABLE 1. Example lunar optical links [9].

to radio frequency (RF) communications, but the corresponding equipment is also better suited for space missions. Lasers have a smaller form factor, weigh less, and require less power than RF antennas. Furthermore, the high directivity of lasers minimizes the need for spectrum regulation. Regarding lunar communications, according to [7], optical systems typically operate at a wavelength of 1550 nm.

NASA has already conducted experiments investigating the strengths and weaknesses of optical communications and is expected to incorporate an optical communications terminal in the Artemis II mission, scheduled for after September 2025. While the RF communication system in the S-Band will support data rates of a couple of Mb/s, the optical terminal is expected to achieve rates up to 260 Mb/s.

Based on technologies tested by NASA, [9] provides data rates considered achievable in the future for various optical links, as shown in Table 1. The last two columns have been added to provide a better perspective. They follow by assuming that the spacecraft (Orion in [9]) is in a distant retrograde orbit 80,000 km from the moon's surface, while the (lunar) relay is in a high circular polar orbit 10,000 km from the moon's surface. For the moon-earth trunk link, a coherent receiver with QPSK modulation is assumed, while the other links use PPM with non-coherent receivers. PPM is well-suited for power-constrained links but requires higher bandwidth, posing a challenge for constrained space-grade electronics, thus limiting achievable data rates. On the other hand, coherent formats and receivers allow for increased data rates and are power-efficient, but currently, they require adaptive optics to mitigate atmospheric distortions. It is important to note that the dependence of optical communications on Earth's weather conditions necessitates the consideration of hybrid systems, in which RF systems provide a robust backup, which is particularly important in emergency scenarios.

LUNAR SURFACE WIRELESS NETWORK

As more and more lunar assets will be deployed in the future, it is necessary to establish an architecture for lunar surface communications. Relay satellites orbiting the moon could be utilized for this purpose. However, this approach becomes less effective as the number of users grows, and so the need for establishing a lunar wireless network arises.

Instead of designing separate standardization efforts, this network could be based on existing cellular technologies standardized by the 3GPP, such as 4G/LTE and 5G, which are mature and have proven to be robust [10]. The aforementioned technologies have been designed for ter-

restrial users, and adapting them to the space environment is not straightforward; a number of problems need to be addressed. Power consumption in space, unlike on Earth, is a principal constraint affecting the design of the equipment. The system also needs to be able to sustain the extreme temperatures and radiation, while the rugged lunar surface with its multiple craters, which need to be explored for valuable resources, can lead to intermittent connectivity. One approach to consider is leveraging the upper layers of the 5G network architecture and tailoring the physical layer for the lunar environment [11].

In the near future, as part of the Artemis program, Nokia Bell Labs will deploy the first 4G/LTE system at the moon's south pole [10]. The system will consist of a 4m-tall base station mounted on a lander, a 1m-tall rover, and a free-flying "hopper," utilizing a 20 MHz channel in Band 3 (1710–1785 MHz uplink, 1805–1880 MHz downlink). Even though this will not be a critical system, it will provide invaluable information for future lunar 3GPP-based systems.

DELAY TOLERANT NETWORKS

Much like the internet protocol (IP) used in terrestrial communication networks, an inter-networking protocol known as the delay/disruption tolerant networking (DTN) bundle protocol has emerged to overcome the challenges specific to cislunar space. DTN was introduced in [12] to address the issues that arise when using typical internet protocols such as TCP and UDP to transmit data over links with extreme propagation delays. As opposed to the datagrams used in IP, DTN uses data bundles, which comprise meta-data such as source-destination identifiers and priority of the bundles. With the use of bundles, DTN employs a *custodial re-transmission* protocol, where a node takes custody of a bundle and allocates enough resources to maintain a copy of the bundle until some other node present further in the communication path takes custody of the bundle. Custodial re-transmission, along with a non-sequential file transfer protocol known as CFDP, enables DTN to provide reliable communication across vast distances.

RECONFIGURABLE INTELLIGENT SURFACES

Reconfigurable intelligent surfaces (RISs) have emerged in recent years as a technology that allows the construction of intelligent propagation environments. The main idea is that a software-controlled surface steers incident signals toward desired directions, effectively acting as a "smart" mirror. This is achieved by partitioning the surface into cells with adjustable properties, which can be altered to change the direction of the reflected signal. In cases where communicating users do not have a clear line-of-sight (LoS) with each other but do have a clear LoS with an RIS, a virtual LoS between the users can be established by suitably reflecting incoming signals, consequently improving the communication quality. Their energy-efficient nature and ease of deployment are additional characteristics that invigorate their potential for use in the lunar surface network.

A study about the use of RISs on the surface of Mars is provided in [5], and it contains many of the insights that hold for the lunar surface as well. Strategically positioned RISs can play an

important role in mitigating intermittent connectivity by establishing virtual LoS links at a low cost. For instance, in a scenario where a rover explores a crater, communicating with other lunar assets outside will be challenging due to the absence of a clear LoS and the elevation difference. By deploying an RIS on the edge of the crater, the connectivity can be improved in a robust and cost-efficient way since orbiters need not be used. Additionally, RISs can also be utilized when communicating users have a clear LoS but the distance between them is considerable. More sophisticated designs for RISs exist where incident signals are also amplified when reflected, thus, in the interest of preserving energy for mission-critical equipment, such variants can be advantageous. Utilizing RISs to enhance the performance of the localization system in scenarios where lunar assets do not have a LoS with orbiting satellites is another use case highlighted in [5]. The development of suitable algorithms constitutes a promising research avenue.

Despite the numerous advantages of RISs, there are some issues that need to be addressed. To effectively reflect signals toward each user, it is essential for the RISs to acquire accurate channel state information (CSI) for each link. However, their passive nature makes this challenging. The manufacturing process is further complicated by the extreme conditions on the surface of the moon, while its distinct topography necessitates detailed studies for optimal placement. Given the early development stage of lunar communication protocols, it is important that the deployment of RISs is carefully planned.

NEW METRICS: SEMANTICS, GOAL/TASK-ORIENTED, TIMELINESS

Another interesting development considered for 6G, which can be impactful in cislunar communications as well, is the set of emerging novel metrics, such as the semantics of communication, goal-oriented and task-oriented communications [3], as well as timelines, freshness, and age of information (AoI) [4]. Traditionally, communication networks have been designed with the mindset of transmitting bits from one point to another and keeping either a high throughput or a low delay in this transmission. In emerging 6G, as well as in cislunar communications, rather than transmitting bits from one point to another, accomplishing a goal could be a more suitable metric. With robots, machines, sensors, and humans collaborating in missions, semantics/context/meaning of transmitted information and the purpose and end-to-end completion of the mission (goal, task-oriented communication) will be critical.

Going beyond the traditional metrics of throughput and delay to more mission-relevant metrics such as age and freshness of information will play important roles. In this regard, AoI, which penalizes the staleness of the updates, will be a crucial performance quantifier that can be leveraged for the design of transmission policies in cislunar communications.

CISLUNAR CHALLENGES AND DESIGN PROBLEMS

In addition to the delay-tolerant networking, adequate cislunar infrastructure must be integrated into the cislunar communication system to provide enough coverage even to the outskirts of the lunar

far side (LFS). Due to tidal locking, LFS does not have a direct visible link with the Earth. The use of a single surface relay or a single orbiting relay may not be sufficient to provide coverage to all the equipment present on the lunar surface. Further, all these lunar surface probes and relays work in a severely energy-constrained environment, which limits their rate of data transmission and hence the freshness of data degrades. Thus, the design of satellite constellations and scheduling policies to achieve timely communication in an energy-constrained environment arises as a fundamental problem when designing the cislunar infrastructure.

LUNAR COVERAGE

The satellite constellations must be designed to maximize the lunar coverage while positioning them in minimal energy orbits (i.e., orbits that require a minimal number of maneuvers to stay in orbit), which in turn minimizes the fuel consumption and maximizes the mission duration of the satellites. Due to the limited orbital stability of conventional lunar orbits, the design of halo orbit constellations has been widely studied in the literature due to the stationarity of the Lagrangian points and the large surface visibility provided by the halo orbits. Halo orbits with higher out-of-plane amplitude have high orbital stability. However, they may decrease the lunar equatorial coverage. On the contrary, DROs, which also exhibit stable orbits, provide sufficient coverage on the lunar equator but lack coverage at the poles. Combining the complementary strengths of the two orbits, [13] introduces the design of hybrid halo-DRO satellite constellations, where they show that a four-halo orbit satellite constellation or a four-DRO satellite constellation can provide total lunar surface coverage. There has been promising research in this direction, where the design of multiple orbit constellations, elliptical inclined orbit constellations, and lunar frozen orbit relays are explored.

TIMELINESS IN COMMUNICATION

In many remote monitoring systems, timely communication is necessary to minimize the drift in estimates. Due to the significant delays present in the cislunar space, it is clear that timely communication is a vital but challenging feature that must be integrated into cislunar communication systems.

The work in [2] is one such attempt, which uses the AoI metric for the design of a satellite constellation to communicate with the LFS in a timely manner. They propose a framework for the design of a hybrid constellation that utilizes both a circular orbital satellite and a halo orbital satellite placed at the Lagrange L2 point, as shown in Fig. 3. The use of a halo orbital satellite provides the coverage to LFS, which does not have direct visibility to the Earth while enabling sensors within the coverage of both the satellites to relay their data through multiple paths. The use of multiple pathways is a simple yet effective design technique to improve the timeliness of such systems, but it comes with the added complexity of out-of-order transmissions. By considering the hybrid constellation to be a multi-source, two-server queuing model, they design policies for scheduling the source transmissions between the two satellites.

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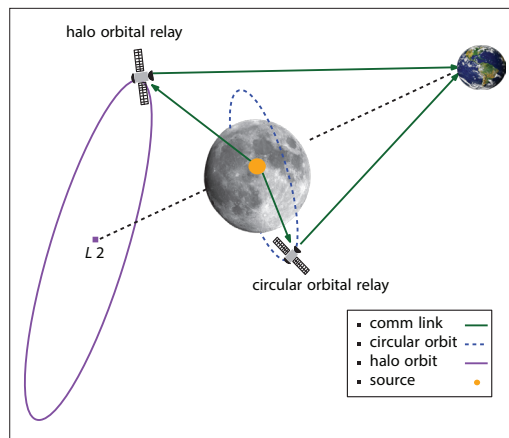


FIGURE 3. Hybrid halo-circular orbital satellite constellation depicting a multi-relay communication setting.

A design challenge that may naturally arise in cislunar communications stems from the mobility of satellites. Unlike their terrestrial counterparts, some satellites, especially the circular orbital satellites, may only be periodically available for some regions on the lunar surface. We believe that such systems can be designed by modeling the satellites as servers that *take vacations* periodically, where appropriate scheduling and sampling policies should be established so as to minimize the AoI. Moreover, when designing scheduling policies to optimize for timeliness, age-agnostic scheduling policies would play a pivotal role in the cislunar space since they do not require an explicit feedback channel. Age-dependent policies, on the other hand, would require an exact replica of the associated age process, which can be difficult to maintain due to the long propagation delays and packet errors and, therefore, should almost always be supplemented with an ARQ-like protocol. Hence, we believe the design of age-agnostic policies such as probabilistic schedulers and cyclic schedulers would play an important role in the future of cislunar communications.

LUNAR NIGHTS AND ENERGY CONSTRAINTS

One of the most challenging aspects in the cislunar space is the lack of a consistent and reliable energy source. The main source of energy for the lunar probes and relays comes in the form of solar energy. Lunar nights, which can last as long as 14 earth days, are one of the greatest obstacles that must be overcome in cislunar space. During lunar nights, the sensors and relays operate with a low energy budget, and hence, the rate of transmission must be reduced. We believe the design of such systems can be modeled as servers whose service rate is a time-dependent function, while the sensors can be modeled as energy-harvesting nodes. Moreover, to reduce the transmissions, we can incorporate semantics, such as the occurrence of an event or the change in the state of the system, to design event-driven transmission policies that can greatly minimize the number of transmissions. In addition to energy constraints, during lunar nights, the sensors may be subjected to extreme temperature changes. Thus, they must be carefully designed to operate under harsh environments.

Due to the long propagation distances, the signal strength of the transmitted data drastically decreases the further it is from the source. This, along with solar interference, can heavily degrade the signal-to-noise ratio (SNR) of the signals at the earth observatories. To mitigate this issue, an ideal choice is to relay the signal via multiple hops. To accomplish this, satellite orbits must be established at multiple altitudes both around the Earth as well as around the moon. Existing low-earth orbit and high-earth orbit satellites can be refurbished, and new lunar orbital satellites must be established for this purpose. However, the establishment of a new orbital satellite must be carefully chosen since the gravitational influence of the sun and Jupiter may disrupt the stability of their orbits. Figure 4 envisions one such multi-hop relay network. In general, amplify-and-forward (AF) is preferred in situations where simplicity, low cost, and low latency are priorities, while decode-and-forward (DF) is chosen for scenarios that demand high reliability and better performance in noisy or poor channel conditions. This is why satellites can AF their packet to ground stations in multi-relay communication [2]. Meanwhile, the specifics of multi-hop relay protocols are still under-explored and are an active area of research. We can expect DF to be used if there are more than a certain number of hops in communications.

CACHING

From an information freshness perspective, cached data is advantageous in minimizing transmission delays. However, if the cached data becomes out-of-date, it can adversely affect the freshness of the information received. Hence, outdated data must be evicted and replaced with fresh data. Focused on this phenomenon, caching has been extensively studied in information dissemination networks where the optimal cache update rates and cache update policies have been developed to maximize information freshness. Thus, these caching algorithms emanating from information dissemination networks would be instrumental for the establishment and proper placement of orbital caches (caching relays) in the cislunar space. We envision that these orbital caches will be the gateway between the earth-moon communication links in the future.

Depending on the altitudes of these caching relays, they may be so expensive that no single company or a single country may be able to afford their launch, or even if they do, they may not be financially sustainable in the long run. Thus, many of these relays may have to be shared among different entities (e.g., different corporations or different countries). Consequently, necessary steps must be taken so as to maintain data privacy for these multi-tenant caches. In this regard, information-theory-based privacy techniques described below can be instrumental.

PRIVATE INFORMATION READ, WRITE, CACHE

Another technology that can facilitate the cooperation of different parties in space missions is private information retrieval (PIR) and its extensions, such as private read-update-write (PRUW) and private multi-party computation. As the term sug-

gests, PIR allows users to download data entries from a shared system of databases without the databases being able to determine which entries were actually desired. This is achieved by sending carefully crafted queries to the different databases. The individual answers from these databases do not leak information, but when combined and processed by the user, they yield the desired data.

Such approaches are easy to implement on the application layer and can be extremely useful in scenarios where different parties share valuable equipment but do not fully trust each other. An important extension of PIR is PRUW which allows users to download, update, and upload the updates back to a storage system without revealing the content downloaded, uploaded or their positions. These features are especially beneficial for securely updating cache entries in shared satellites without revealing the details of the updates.

The main obstacle in adopting these technologies is the requirement of having multiple databases with a set of common entries that the user can query. Given the limited assets in space, storing data across multiple databases may currently be a challenge. However, as our lunar and cislunar infrastructure improves, these methods could become prime candidates for enhancing security and trust-free cooperation. This also points to the need to study single-database PIR further.

SECURE COMMUNICATION AND STORAGE

Different entities in the cislunar space are expected to cooperate and form networks, as discussed previously, with secure communications being the backbone of this synergy. The subject of security is rather broad but in this section we view it through the lens of wireless communications and information theory.

INTERFERENCE DETECTION IN SPACE

Cislunar communication is increasingly challenged by interference, which can disrupt signals between Earth and lunar missions. The growing number of satellites and communication systems in both Earth orbit and cislunar space amplifies the risk of electromagnetic interference, co-channel interference, and adjacent-channel interference. These forms of interference can degrade the quality of communication links essential for supporting astronauts, rovers, and other lunar operations. Moreover, the unique environmental conditions of cislunar space, including temperature fluctuations and varying orbital dynamics, further complicate communication systems, making them susceptible to unexpected anomalies. To mitigate these issues, a robust regulatory framework for spectrum allocation is crucial to ensure seamless operations across diverse missions while addressing potential malicious interference attacks such as jamming.

In the absence of adequate regulatory oversight and given the difficulties of establishing it, advanced radio frequency interference (RFI) detection techniques specifically tailored for the cislunar environment must be developed. For example, [14] proposes machine learning (ML) methods for RFI detection, leveraging synthetic data that account for physical phenomena such as the effect of the lunar surface's brightness temperature on signal noise. Additionally, a proof of concept in [15] demonstrates the fea-

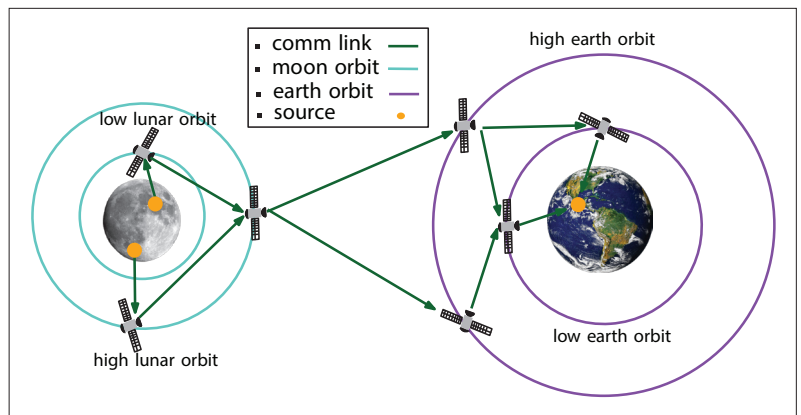


FIGURE 4. Multi-hop satellite relay network connecting low and high orbit satellites to improve the quality of transmissions in cislunar space.

sibility of deploying ML-based RFI detection in the resource-constrained environment of space, paving the way for further research on the topic.

PHYSICAL LAYER SECURITY

Designing new systems for space communication, though challenging, offers an opportunity to adopt innovative concepts that are fundamentally different from the current approaches. Instead of solely relying on cryptographic schemes for secure communications, which depend on the limited computational capabilities of adversaries and may not be fully compatible with the vision for long-term secure space communications, it is worth considering information-theoretic approaches where the requirement for secure communication is incorporated in the design of the physical layer.

The main idea behind physical layer security is to guarantee that the observations obtained by an eavesdropper and the secret message of the transmitter are statistically independent; with this constraint in place, computational power becomes irrelevant. As discussed in [6], multiple transmitters can enhance the rate of secure communication by means of cooperative jamming and structured signaling. This idea extends to scenarios that include untrusted relay nodes. Given the high cost of deploying relay satellites in lunar orbit, adopting such technologies for secure communications, where relays are treated as untrusted nodes, can facilitate cooperation between different space agencies and companies.

While promising, existing results come at the expense of reduced transmission rates, which can be improved by use of multiple antennas or RISs; and are mostly information-theoretic in nature with some polar-code-based implementations, which can be made more practical with design of explicit codes. It is crucial to note that lessons learned in space, where stakes are high, can facilitate the development of analogous terrestrial technologies.

SECURITY AND SCHEDULING

In cislunar space, the aspect of secure communications can significantly impact scheduling policies. By monitoring the transmission patterns of relay satellites, adversaries can obtain valuable information. For instance, unusually long periods of communication may indicate significant events, such as emergencies. By leveraging that, adversar-

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ies can choose when to launch an attack so as to increase the incurred damage. Given the critical role of relay satellites, their periodic coverage of areas, and their limited resources, developing transmission and caching policies that also take security into account is an interesting research direction.

CONCLUSION

This article aims to act as a catalyst for the engagement of the research community with the problems arising in cislunar communications. Toward that end, we discussed cislunar specific communication challenges and investigated how existing terrestrial communication technologies can be relevant in shaping the future of cislunar communications. We examined how novel metrics, such as AoI, can be used to design satellite constellations and transmission policies. With privacy being a major concern, we discussed the strengths and weaknesses of physical layer security and private information retrieval.

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